DATA SCIENCE WORK SAMPLE / ENERGY

PSCo Hourly Demand Analysis

A focused Python and SQL case study using public EIA-930 balancing-authority data to assess hourly demand quality, operating patterns, and forecast error.

SOURCE ROWS SCANNED	PUBLIC SOURCE FILES	PSCO HOURLY ROWS	ONE-HOUR MODEL MAE
3.85M	**14 / 616 MB**	**61,368**	**103.72 MW**

Business question

How can reported hourly demand data be validated and analyzed to surface useful operating patterns and evaluate short-term forecast baselines?

What I built

EIA-930 bulk CSV	Python chunked ingest	SQLite + SQL	Analysis + dashboard

Across 14 public EIA files for 2019-2025, the pipeline scanned 3,848,038 multi-authority rows (616 MB) in 50,000-row chunks and retained 61,368 PSCo hourly records. It stores the selected table in SQLite, records quality checks, and runs SQL for peak hours, demand profiles, and forecast error.

Skills demonstrated

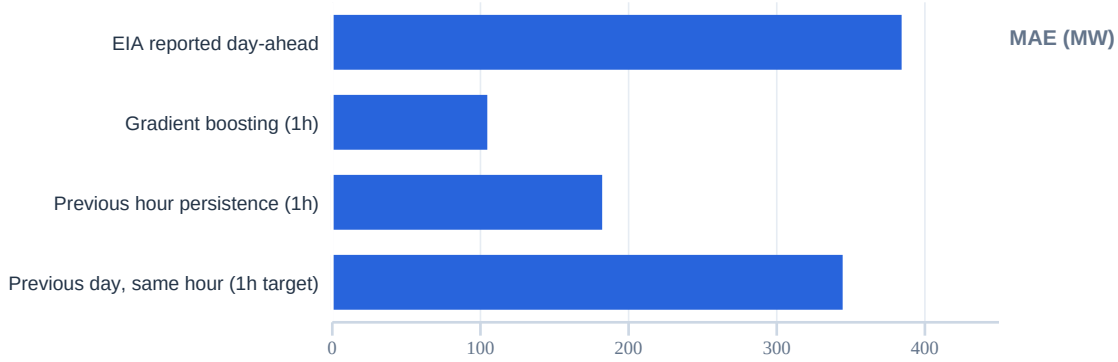
Python / data engineering	Chunked CSV ingestion, type parsing, timestamp checks, invalid-value handling
SQL / database	SQLite table, time index, documented aggregation and quality queries
Statistics / ML	Chronological holdout, persistence baselines, MAE and RMSE
Communication	Demand dashboard and concise operational interpretation

Scale note: the measured workload is the multi-authority source scan; the analysis table is the PSCo-only extract. The raw CSVs are public and are downloaded by the reproducible pipeline.

MEASURED RESULTS

Forecasts, checked honestly

Results below are transcribed from the candidate's successful 2019-2025 PSCo pipeline run. The 80/20 split is chronological; the held-out period contains 12,233 hourly observations. The repo includes commands to reproduce the analysis from EIA's public bulk files.



On this rolling one-hour-ahead holdout, gradient boosting's MAE is 42.8% lower than previous-hour persistence and 69.8% lower than the previous-day baseline. Results are encouraging for this evaluation window, not proof of operational performance.

Evaluation	MAE (MW)	RMSE (MW)	Forecast horizon
Previous-hour persistence	181.26	230.17	1 hour
Gradient boosting	103.72	146.03	1 hour
Previous-day same hour	343.59	455.42	1 hour target
EIA reported forecast	383.39	481.10	Day-ahead

Forecast horizons differ: the EIA metric is day-ahead; the model and persistence baselines are rolling one-hour-ahead. The EIA number is shown for context and is not an apples-to-apples ranking.

Data quality

The run had 61,368 PSCo rows, zero duplicate timestamps, and zero missing hourly slots. It flagged 24 missing demand values and 13 nonpositive values; all 37 were excluded from demand analysis while preserving the raw observations.

Scope and limits

EIA-930 is balancing-authority data, not customer-level consumption. Timestamps are UTC end-of-hour. This project analyzes demand and forecasts only; it does not assess generation mix, weather effects, causation, or grid dispatch. Findings support exploratory analysis, not operational decisions.

Source: U.S. Energy Information Administration, Form EIA-930 public bulk CSVs, eia.gov/electricity/gridmonitor. Portfolio: portfolio-website-vignesh--kvk1011.replit.app. Source code, SQL queries, tests, and dashboard are in the accompanying project archive.